

The Neurophysiology of Cybersickness in Virtual Reality: A Systematic Review of Neural Correlates and Neurostimulation

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Abstract—Cybersickness remains a barrier to virtual reality (VR) adoption, and subjective questionnaires alone are insufficient for system-level diagnosis of its determinants. This systematic review, conducted following the PRISMA 2020 statement, examines neuromarkers of cybersickness in VR: signals derived from non-invasive recording or modulation of brain activity that index or modulate the user’s state. A search of PubMed, Web of Science, and Scopus up to 27 March 2025 identified 92 eligible studies: 77 using electroencephalography, 4 using functional near-infrared spectroscopy, and 11 using neurostimulation (5 galvanic vestibular stimulation, 3 transcranial direct-current stimulation, 2 transcranial alternating-current stimulation, 1 other). No eligible studies using fMRI, MEG, or PET were retrieved. Across electroencephalography studies, cybersickness is associated with increased low-frequency power (delta, theta) and decreased alpha-band power, with topographies varying by paradigm and hardware. At the corpus level, VR display hardware significantly modulates band-power responses (alpha: $p = 0.0036$, large effect; beta: $p = 0.0146$), indicating that prior contradictions in the literature are partly attributable to hardware confounds rather than to differences in cybersickness per se. Egger’s test detects publication bias across the corpus ($p = 0.011$). A methodological audit identifies the prevalence of accuracy-only reporting in classification studies, weak baseline choices, and ad hoc sick-versus-non-sick thresholds. Relative to the most recent prior scoping review [1], which covered 33 EEG-only studies up to September 2021, this review covers a larger and more recent corpus, explicitly includes neurostimulation alongside neuroimaging, and contributes corpus-level meta-analytic findings. We outline reporting recommendations and identify closed-loop modulation and individual susceptibility prediction as directions where the existing corpus already provides actionable starting points.

Index Terms—Human-centered computing, Human computer interaction (HCI), Interaction paradigms, Virtual reality.

I. INTRODUCTION

Virtual Reality (VR) technology has permeated numerous sectors beyond entertainment, becoming integral tools in fields such as healthcare, professional training, and education [2]. Generally, VR is defined as a computer-generated simulation that enables users to perceive and interact with a three-dimensional environment through real-time multisensory feedback. A typical VR system includes a head-mounted display for stereoscopic rendering, motion-tracked controllers or sensors for user input, and a computing unit for environment simulation and rendering. However, there is no unifying definition of VR [3].

A barrier to widespread VR adoption is the onset of cybersickness (CS) [4]. Cybersickness is a form of visually induced

motion sickness (VIMS) [5] characterized by symptoms such as nausea, dizziness, and disorientation that arise during or after immersion in a virtual environment [4]. The relationship between cybersickness, VIMS, simulator sickness, and motion sickness is treated in detail in Section II; the present review uses VIMS as the umbrella term when introducing constructs and uses the term each cited study employs when reporting its findings. Cybersickness affects a substantial portion of the user population, with studies reporting prevalence between 60% and 95% [4], degrades user experience, impairs task performance, and can lead to premature session termination [6].

Assessment of cybersickness has historically relied on subjective post-hoc questionnaires such as the Simulator Sickness Questionnaire [7]. Such instruments are retrospective, prone to recall bias, and do not capture the dynamic nature of the experience [1], [8]. This limitation creates a need for objective measures grounded in the user’s physiology and for interventions that can modulate the user’s state during exposure rather than after the fact.

Two converging lines of research have responded by examining the brain. The first records neural activity — primarily electroencephalography (EEG), and to a lesser extent functional near-infrared spectroscopy (fNIRS) — to identify signals that index the onset, severity, or trajectory of cybersickness. The second applies non-invasive brain stimulation, including galvanic vestibular stimulation (GVS), transcranial direct-current stimulation (tDCS), and transcranial alternating-current stimulation (tACS), to modulate the underlying processes that produce sickness. We use the term *neuromarkers* to refer to signals derived from either non-invasive recording or non-invasive modulation of brain activity in this context. For a VR system designer, neuromarkers carry information that subjective questionnaires cannot: continuous, time-resolved indices of user state that can in principle drive adaptive content, comfort thresholds, and closed-loop mitigation.

Chang et al. [1] synthesized 33 EEG studies published before September 2021 in a scoping review and characterized the EEG analysis pipeline used in cybersickness research. The present systematic review differs in four respects. First, the scope extends beyond EEG to include functional near-infrared spectroscopy and non-invasive neurostimulation, consistent with the integrated frame of neuromarkers. Second, the corpus is larger and more recent: 92 studies retrieved up to 27 March 2025 from three curated databases (PubMed, Web of Science, Scopus), compared with 33 studies retrieved from

Web of Science and Google Scholar up to September 2021. Third, the synthesis includes corpus-level inferential statistics on hardware and methodological confounders, in addition to the pipeline characterization shared with prior work. Fourth, this review situates the empirical findings within the broader context of the field: the theoretical frameworks invoked to explain cybersickness, the terminology used across the cybersickness, VIMS, simulator sickness, and motion sickness literatures, and a critical appraisal of recurring methodological shortcomings. Where the prior scoping review concentrates on findings and pipelines, this review additionally examines the conceptual and methodological scaffolding in which those findings are produced.

The contributions of this review are threefold. First, we provide a systematic synthesis of cybersickness neuromarkers in VR across 92 studies, covering neuroimaging and neurostimulation under a unified frame. Second, we report meta-analytic evidence that VR display hardware significantly modulates the band-power response associated with cybersickness ($p = 0.0036$ on alpha, large effect; $p = 0.0146$ on beta), recasting prior contradictions in the EEG–cybersickness literature as partly attributable to hardware confounds rather than to differences in cybersickness per se. Third, we conduct a methodological audit of the field, including detection of publication bias (Egger’s test, $p = 0.011$) and examination of reporting practices in classification studies, and derive concrete recommendations for future VR-EEG and VR-stimulation work.

The remainder of this paper is organized as follows. Section II reviews the relevant theoretical frameworks for cybersickness and the principal neuroimaging and stimulation modalities. Section III describes the review methodology and search strategy. Section IV presents the corpus, the per-modality synthesis, and the corpus-level meta-analytic findings. Section V discusses methodological challenges and outlines future directions. Section VI concludes.

II. THEORETICAL AND METHODOLOGICAL BACKGROUND

A. Terminology and Etiological Frameworks

Before surveying the theories proposed to explain cybersickness, it is useful to delimit the construct itself and to distinguish it from related forms of sickness and from perceptual phenomena that often co-occur with it.

Motion sickness is the broadest construct, encompassing adverse responses to both physical and visually specified motion [9], [10]. Within it, visually induced motion sickness (VIMS) designates sickness elicited primarily by visual motion, including cases where physical body motion is absent or inconsistent with the visual scene [11], [12]. Cybersickness is most often treated as the VR-specific instance of VIMS; simulator sickness, gaming sickness, and related labels denote overlapping exposure contexts rather than distinct symptom categories [13]–[15]. This review adopts the above hierarchy when introducing constructs, but reports each included study using the terminology its authors chose, in order to avoid imposing a uniform vocabulary on a literature that does not yet share one.

Symptom measures in this literature must also be distinguished from related perceptual phenomena. The dominant instrument is the Simulator Sickness Questionnaire (SSQ), which yields nausea, oculomotor, and disorientation subscales [7]; it is frequently complemented by the single-item Fast Motion Sickness Scale (FMS) for continuous, in-task ratings [16]. Broader diagnostic criteria for motion sickness and VIMS extend these subscales to gastrointestinal, thermoregulatory, arousal, dizziness or vertigo, headache, and visual or oculomotor domains [12]. Distinct from these symptom measures,vection denotes the visually induced illusion of self-motion: it frequently covaries with sickness and figures in several etiological theories, but is not itself a symptom category [17], [18].

Several etiological frameworks have been proposed to explain cybersickness and the wider VIMS phenomenon; they are not mutually exclusive and often emphasise complementary mechanisms.

The most influential of these is *sensory conflict theory*, which attributes sickness to inconsistency between visual, vestibular, and proprioceptive signals, or between current sensory input and the prior exposure history encoded by the observer [19]–[21]. Two refinements within this family further specify the nature of the conflict: subjective vertical mismatch frames it as a discrepancy between sensed and expected gravitational vertical [22], while rest-frame conflict locates it in the observer’s difficulty identifying a stable spatial reference [23]. This family is particularly pertinent to VR, where HMDs routinely present visual self-motion while vestibular and proprioceptive signals indicate little or no corresponding physical movement.

A second framework, *postural instability theory*, locates the cause of sickness not in sensory mismatch but in a sustained inability to maintain postural control under unfamiliar sensorimotor demands [24], [25]. The theory is most directly testable in VR paradigms involving standing, walking, or large-field optic flow, where body sway and control dynamics can be measured objectively and have been reported to precede or accompany subjective sickness [26]. A related strand of recent VR work returns to the role ofvection itself, showing that sickness intensifies when visually induced self-motion violates the user’s expected motion or control state [27], [28];vection is treated here as a perceptual condition that may contribute to sickness rather than as a separate theory of motion sickness in its own right.

A further set of proposals address narrower mechanisms or different levels of explanation. Vergence-accommodation conflict identifies a mismatch between binocular convergence and the fixed focal distance of stereoscopic displays as a source of oculomotor discomfort [29], while oculomotor theories more broadly relate sickness to specific eye-movement patterns, such as nystagmus or ocular torsion, evoked by the visual stimulus [30]. At an evolutionary level, Treisman’s toxin-defense hypothesis explains why disturbed sensory integration should give rise to nausea and emesis, but does not specify the proximate sensory conditions that trigger them [31]. Finally, predictive-coding formulations recast sensory mismatch as a persistent prediction error between expected and observed

input, offering a unifying computational vocabulary for the preceding theories [32].

B. Neural Recording and Modulation Modalities

Cybersickness has historically been assessed through subjective reports, but a growing body of work uses neural recordings, and more recently non-invasive brain stimulation, to characterise the user's state during VR exposure or to actively modulate it.

Electroencephalography (EEG) is the most widely used recording modality in this literature. Its high temporal resolution supports the study of spectral power, event-related potentials, and connectivity dynamics throughout the development of sickness, and its portability makes it compatible with HMD-based and other interactive VR paradigms [26], [33]. These strengths are offset by limited spatial specificity, particularly for deep vestibular and autonomic structures, and by methodological sensitivities, such as motion artifacts, electrode density, reference choice, and baseline selection, which constrain the interpretation of reported effects.

Functional near-infrared spectroscopy (fNIRS) is an emerging optical technique that measures cortical hemodynamic changes via near-infrared light [34], [35]. It offers an intermediate trade-off between temporal and spatial resolution and tolerates moderate movement, making it well suited to VR paradigms such as HMD-based roller-coaster exposure, VR locomotion, or stereoscopic viewing [36]–[39]. Because it relies on hemodynamic rather than electrical signals, its temporal resolution remains substantially lower than that of EEG, and its optical penetration restricts measurement to superficial cortical regions.

Beyond recording, non-invasive neurostimulation has begun to be used to modulate sickness-relevant processes during VR or visually induced motion exposure. Reported approaches include galvanic vestibular stimulation, transcranial direct-current stimulation, transcranial alternating-current stimulation, and related techniques [40]. By directly perturbing vestibular or cortical activity, these methods provide a causal complement to the correlational evidence offered by EEG and fNIRS.

III. REVIEW METHODOLOGY

A. Search Strategy

A systematic literature search was conducted across three databases: PubMed, Web of Science, and Scopus. A publication-date cutoff of 27 March 2025 was applied to PubMed and Web of Science; no earlier date restriction was imposed, so the search captures the full historical record of the field. The query targeted the intersection of three concept blocks: brain-directed measurement or modulation methods, cybersickness and related sickness constructs, and immersive virtual reality. The full query was:

```
((("Brain Computer Interface" OR BCI OR EEG OR
electroencephalogram OR ERP OR MEG OR
magnetoencephalography OR fMRI OR ECoG OR
fNIRS OR PET OR "positron emission
tomography" OR neurostimulation OR "brain
```

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stimulation" OR "non-invasive brain
stimulation" OR tACS OR "transcranial
alternating current stimulation" OR tDCS
OR "transcranial direct current
stimulation" OR TMS OR "transcranial
magnetic stimulation" OR "galvanic
vestibular stimulation" OR GVS)
AND (Cybersickness OR "simulator sickness" OR
"visually induced motion sickness" OR VIMS
OR "motion sickness")
AND ("virtual reality" OR VR OR "head-mounted
display" OR HMD OR "immersive environment
"))
```

B. Study Selection and Eligibility Criteria

Records identified by the search were screened in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement; the full selection process is summarised in the PRISMA flow diagram (Figure 1). The combined search returned 443 records: 57 from PubMed, 147 from Web of Science, and 239 from Scopus. Duplicate removal across the three sources reduced this set to 219 unique records.

Each of the 219 unique records underwent manual title-and-abstract screening against the following eligibility criteria. A record was included if it met all of the following:

- The study employed a neuroimaging modality (EEG, ERP, fNIRS, fMRI, MEG, ECoG, PET) or a non-invasive brain-stimulation modality (tDCS, tACS, TMS, GVS) to measure or modulate brain activity.
- The study addressed cybersickness or a closely related construct (visually induced motion sickness, simulator sickness, or motion sickness in a virtual environment) within a virtual reality or otherwise immersive context.
- The study reported original empirical data from human participants.

A record was excluded if it met any of the following:

- The work was non-empirical (literature reviews, meta-analyses, editorials, letters, or conference abstracts without an accompanying full paper).
- The study did not address cybersickness or a related sickness construct as a primary outcome.
- The full text could not be retrieved.

Title-and-abstract screening excluded 119 records, leaving 100 reports sought for full-text retrieval. Of these, 8 could not be obtained as full text and were excluded at the retrieval stage. The remaining **92 articles** comprise the final corpus analysed in this review.

C. Modalities and Dataset-Reuse Studies

Two aspects of the corpus warrant explicit comment before the synthesis: the modalities that the query covered but that returned no eligible studies, and the inclusion of studies that rely on previously released datasets.

The query included PET, MEG, and fMRI terms alongside the more represented modalities. PET returned no eligible studies; MEG returned a single record, which was excluded at the title-and-abstract stage as not addressing cybersickness or

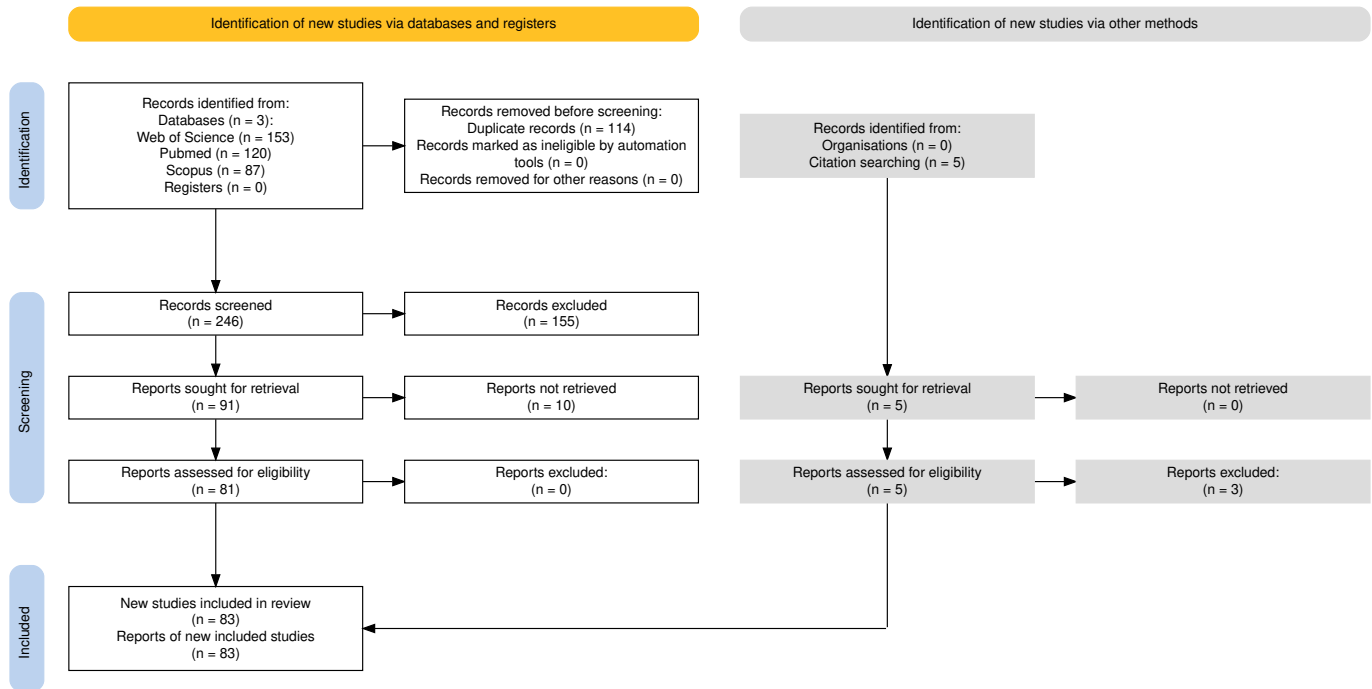


Fig. 1: PRISMA flow diagram for the present review. **Figure to be regenerated to match the updated counts in the text (443 identified; 219 after duplicate removal; 119 excluded at title/abstract; 8 reports not retrieved; 92 included).**^[TODO:prisma:figure]

a related sickness construct; fMRI returned no records at all. The absence of these modalities from the final corpus therefore reflects the current literature rather than a scope restriction imposed by the search.

A subset of retrieved studies train machine-learning classifiers on previously released datasets without recording new data. Pure-dataset benchmarking is a related but methodologically distinct literature, since its primary objective is classification performance rather than the identification or modulation of neuromarkers. Six papers in the final corpus reuse one or more publicly released datasets (in two cases combined with newly recorded data); they were retained because they satisfied all eligibility criteria and are flagged where relevant in the synthesis.

D. Data Extraction and Thematic Synthesis

For each of the 92 included articles, a structured data extraction process was performed. This process was guided by a predefined protocol designed to ensure consistency and to facilitate subsequent comparison and synthesis. The following data points were extracted from each paper:

- 1) **Summary of Contribution:** A concise summary of the study's primary objective and its main contribution.
- 2) **Sickness Measurement:** The methods used to measure or label cybersickness, distinguishing between subjective questionnaires (e.g., SSQ, FMS, VRSQ) and objective physiological or behavioural metrics (e.g., galvanic skin response, postural sway).
- 3) **Methodological Pipeline:** A breakdown of the neurophysiological pipeline, including: (a) the measurement or stimulation modality, (b) the key feature extraction

techniques (e.g., power spectral density, event-related potentials, functional connectivity), and (c) the primary analysis or classification methods (e.g., statistical tests, machine-learning models).

- 4) **Theoretical Framework:** The explicitly stated or implicitly assumed theory of cybersickness (e.g., sensory conflict, postural instability) that guided the study's design or interpretation.
- 5) **Quantitative Performance:** Reported performance measures, such as classification accuracy, correlation coefficients, root-mean-square error, or statistical significance values.
- 6) **Empirical Study Details:** For empirical studies, specific details of the experimental protocol, including the VR environment, display hardware, sickness-inducing stimulus, and number of participants.
- 7) **Reported Shortcomings:** Limitations, caveats, or weaknesses identified by the authors or apparent from the study's design.

Following extraction, each paper was assigned to a primary research-objective category to support quantitative synthesis. An initial set of categories was defined *a priori* based on established research paradigms and refined iteratively during extraction. The final categories, which structure the Results section, are: Classification (Binary, Likert, Continuous), Correlates (Brain Regions, Rhythms), Mitigation (A/B Testing, Real-time, Neurostimulation), User Differences, Dataset Provision, Recovery, and studies on the Effects of Cybersickness on other cognitive or performance metrics.

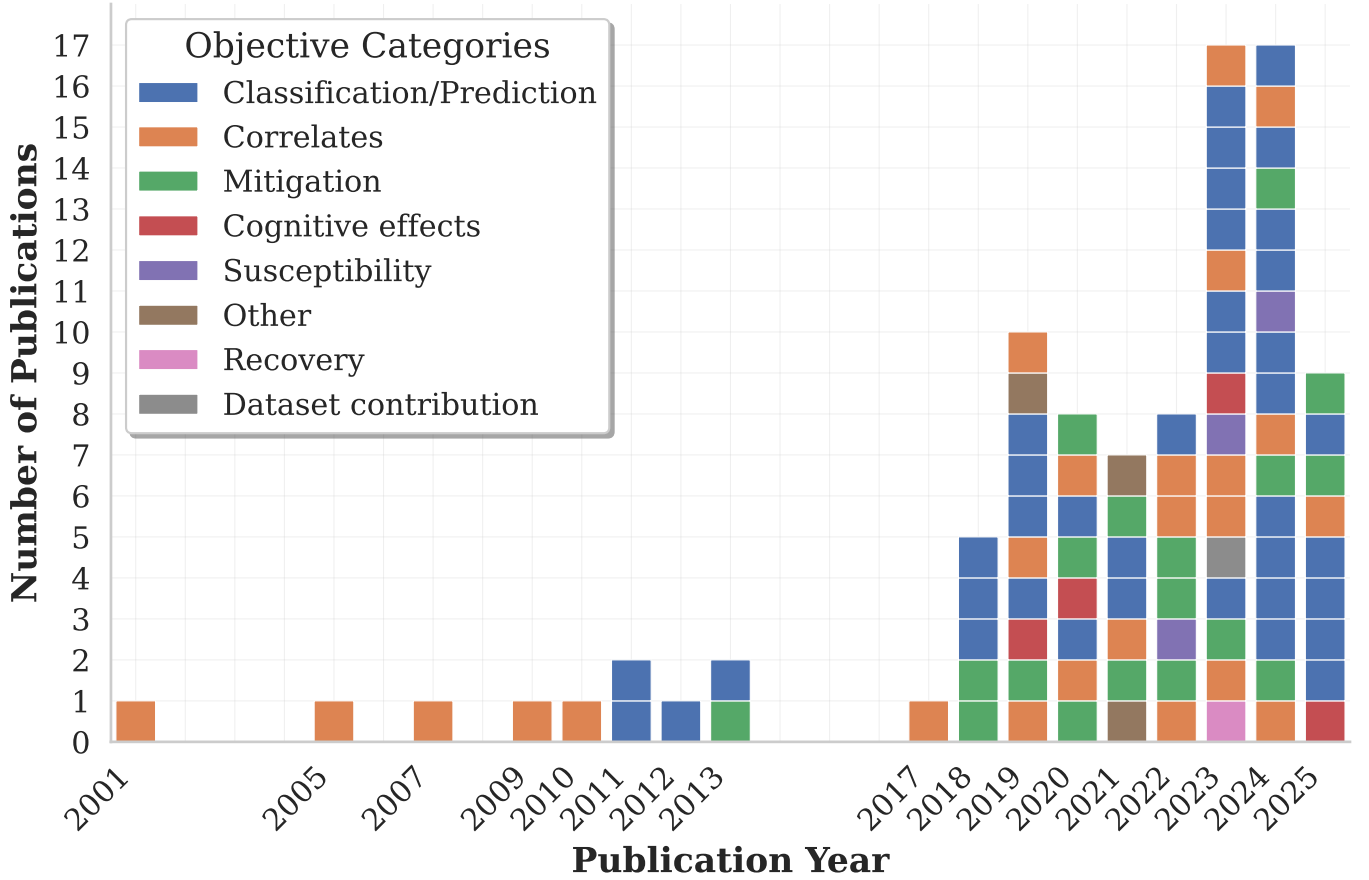


Fig. 2: Distribution of reviewed papers by year and primary research objective.

IV. RESULTS

The 92 corpus studies address four main research objectives: classification or prediction of cybersickness from physiological signals (38 studies), identification of neural and physiological correlates (24), development of mitigation strategies (18), and characterisation of individual susceptibility (3). A residual nine studies have primary objectives in cognitive effects (4), recovery (1), dataset contributions (1), or other framings (3); their corpus-relevant findings are integrated into the relevant subsections, and their full discussion is deferred to section V. This section adds two cross-cutting layers on top of these objectives: subsection IV-A describes the corpus as a whole, and subsection IV-F reports meta-analytic observations across the 92 studies.

Activity in the field has accelerated in recent years: 63.0% of the corpus (58 of 92 studies) was published from 2021 onwards, with a corresponding shift toward classification, mitigation, and susceptibility objectives (Figure 2). Per-study sample sizes have also increased over the same period (Mann–Whitney U on Recent versus Earlier studies, Cliff’s $\delta = -0.308$, $p = 0.022$, medium effect).

Table I lists all 92 studies with their modality, sample size, virtual reality hardware, stimulus paradigm, cybersickness measure, primary objective, and topic. The remainder of this section synthesises findings thematically along the axes above.

TABLE I: All 92 studies included in this systematic review. Modality: EEG, fNIRS, fMRI = neuroimaging; GVS = galvanic vestibular stimulation; tDCS/tACS = transcranial direct/alternating current stimulation; ECG, GSR, EOG, EMG, EGG = peripheral physiology; posturography = balance / centre-of-pressure recording. CS measure abbreviations: SSQ = Simulator Sickness Questionnaire; VRSQ = VR Sickness Questionnaire; FMS = Fast Motion-Sickness Scale; MSSQ = Motion Sickness Susceptibility Questionnaire; cont. = continuous self-report; Likert = single-item Likert; custom = study-specific scale; behav. = behavioural measure only; none = no cybersickness measure reported. Sample size N gives subjects analysed. Topic gives a short noun phrase describing the focus of each paper.

Ref.	Modality	N	VR device	Stimulus	CS measure	Objective	Topic
[41]	EEG	2	HMD	Mixed	custom	Mitigation	EEG-informed scene-simplification mitigation
[42]	EEG+EOG	17	HMD	Driving (passive)	SSQ	Correlates	Alpha connectivity and P3 in susceptibility
[43]	EEG+EOG	33	Screen	Passive viewing	FMS	Correlates	EEG time-frequency correlates of VIMS severity
[44]	EEG	31	HMD	Mixed	SSQ	Correlates	Spectral EEG correlates of VIMS in HMD
[45]	EEG+ECG+GSR+EOG	27	HMD	Active navigation	custom	Classification	Spiking-network EEG+HRV cybersickness prediction
[40]	tACS+GSR	37	HMD	Passive viewing	MSSQ	Mitigation	10 Hz tACS over vestibular cortex
[46]	EEG	68	HMD	Mixed	SSQ	Other	Sex differences in EEG neurofeedback sessions
[47]	EEG+tDCS+ECG+EOG	41	HMD	Passive viewing	SSQ	Mitigation	Placebo tDCS in EEG neurofeedback for cybersickness
[48]	EEG+ECG	39	HMD	Active navigation	SSQ	Cognition	VR-parameter manipulation during EEG neurofeedback
[49]	EEG	15	HMD	Passive viewing	SSQ	Classification	Resting-state EEG entropy classification
[50]	EEG	12	Screen	Passive viewing	SSQ	Mitigation	Rest-frame mitigation with EEG correlates
[51]	EEG+ECG+EOG	20	HMD	Passive viewing	SSQ	Correlates	Heartbeat-evoked potentials as cybersickness correlates
[52]	EEG	19	Projector	Driving (passive)	cont.	Correlates	ICA-based EEG component clusters and sources
[53]	EEG	19	Projector	Driving (passive)	custom	Correlates	Continuous EEG dynamics during driving sickness
[54]	EEG	20	HMD	Other	SSQ	Correlates	ERPs of blur-induced VR visual discomfort
[55]	EEG	16	HMD	Active navigation	SSQ	Correlates	EEG during mobile-VR sickness while walking
[56]	EEG	32	HMD	Mixed	SSQ	Dataset	EEG cybersickness dataset and benchmark
[57]	EEG	32	HMD	Active navigation	SSQ	Classification	EEG-based comparison of VR locomotion techniques
[8]	EEG+ECG+EOG+EGG+ posturography	20	HMD	Active navigation	Likert	Classification	Multimodal EEG+physiology severity classification
[58]	EEG+ECG+GSR	25	Other	Passive viewing	SSQ	Classification	Convolutional-transformer EEG cybersickness detection
[59]	EEG	23	HMD	Passive viewing	SSQ	Classification	Multi-domain EEG feature comparison
[60]	EMG	24	HMD	Passive viewing	MSSQ	Correlates	VEMP changes after VR vection exposure
[61]	GVS	47	HMD	Passive viewing	SSQ	Mitigation	Omnidirectional GVS for 360° video viewing
[62]	EEG+EOG	23	HMD	Passive viewing	SSQ	Classification	LSTM regression of continuous cybersickness severity
[63]	EEG	15	HMD	Mixed	SSQ	Classification	Multi-scale EEG brain-network classification
[64]	EEG	15	HMD	Passive viewing	SSQ	Classification	Bilateral asymmetry features for VRMS detection
[65]	EEG+EOG	22	HMD	Passive viewing	SSQ	Classification	Siam-LSTM with EEG functional brain networks
[66]	EEG+posturography	22	HMD	Active navigation	VRSQ	Classification	Dual-pathway raw EEG and network model
[67]	EEG	40	Screen	Passive viewing	SSQ	Indiv. diff.	EEG band-power correlates of cybersickness susceptibility
[68]	EEG	24	HMD	Passive viewing	cont.	Classification	Deep-learning EEG classification of cybersickness
[69]	EEG	9	Screen	Active navigation	SSQ	Other	Gamma-band EEG visualisation of sickness onset
[70]	EEG+ECG+GSR+EOG	45	CAVE	Active navigation	SSQ	Correlates	EEG and autonomic correlates during navigation
[71]	EEG+ECG+GSR+EOG+ EGG	57	Projector	Active navigation	SSQ	Correlates	Multimodal physiology of navigation cybersickness
[72]	EEG	202	HMD	Passive viewing	Likert	Classification	EEG-to-video transfer for cybersickness prediction
[73]	EEG+EOG	30	HMD	Passive viewing	SSQ	Correlates	Source-localised alpha correlates after VR
[74]	EEG	16	Projector	Driving (passive)	cont.	Classification	ICA-EEG regression of motion-sickness level
[75]	EEG	6	CAVE	Driving (passive)	cont.	Classification	Genetic feature selection for EEG classification
[33]	EEG	43	HMD	Passive viewing	SSQ	Correlates	EEG band-power correlates with continuous SSQ
[76]	EEG+ECG+GSR	40	HMD; Screen	Passive viewing	SSQ	Classification	EEG and physiology fusion for SSQ estimation
[77]	EEG+EOG+posturography	20	Projector	Driving (passive)	cont.	Classification	Multimodal ensemble for VIMS severity
[78]	EEG	18	HMD	Passive viewing	behav.	Classification	Wavelet-packet EEG features for VRMS
[79]	tDCS	20	HMD	Passive viewing	cont.	Mitigation	Cathodal parietal tDCS for cybersickness
[80]	EEG+EOG	22	HMD	Driving (passive)	FMS	Classification	EEG vs. autonomic features for VR sickness
[81]	EEG+EOG	20	HMD	Passive viewing	FMS	Indiv. diff.	EEG biomarker of VRMS resistance

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Ref.	Modality	N	VR device	Stimulus	CS measure	Objective	Topic
[82]	EEG	39	HMD	Mixed	FMS	Classification	Resting EEG prediction of in-car sickness
[83]	EEG+tACS	18	HMD	Mixed	FMS	Mitigation	Online 18 Hz tACS over P3
[84]	EEG	130	HMD	Passive viewing	SSQ	Classification	LSTM prediction of impending cybersickness
[85]	EEG	16	TV	Passive viewing	SSQ	Other	Test-retest reliability of EEG sickness markers
[86]	EEG	9	Projector	Driving (passive)	custom	Correlates	EEG dynamics during VR motion sickness
[87]	EEG	6	CAVE	Driving (passive)	cont.	Classification	BCI estimation of motion-sickness level
[88]	EEG+ECG	25	HMD	Passive viewing	cont.	Classification	Real-time EEG+ECG cybersickness warning
[89]	EEG	9	CAVE	Driving (active)	cont.	Correlates	Wearable EEG and cardiovascular correlates
[90]	EEG	8	Screen	Driving (active)	SSQ	Correlates	Low-channel EEG markers for VIMS
[91]	EEG	36	HMD	Passive viewing	SSQ	Classification	CNN-ECA-LSTM continuous cybersickness regression
[92]	EEG	36	HMD	Passive viewing	SSQ	Classification	EEG spectral and network regression of severity
[93]	EEG	25	HMD	Passive viewing	SSQ	Classification	Wavelet-packet GRU for VR motion sickness
[94]	EEG	9	Screen	Active navigation	SSQ	Classification	Time-domain EEG features for cybersickness
[6]	EEG+EOG	29	HMD	Passive viewing	SSQ	Cognition	P3b ERP attention markers during VR sickness
[95]	EEG+ECG+EOG	8	HMD	Active navigation	custom	Classification	Multimodal symptom-wise cybersickness markers
[96]	EEG	18	HMD	Passive viewing	cont.	Correlates	EEG correlates of display-parameter discomfort
[97]	EEG	40	Screen	Passive viewing	SSQ	Correlates	EEG microstates and functional connectivity
[98]	ECG+GSR	13	HMD	Mixed	VRSQ	Classification	Symbolic ML on peripheral biosignals
[32]	EEG	14	HMD	Passive viewing	SSQ	Correlates	Effective connectivity and predictive coding
[99]	EEG	33	HMD	Passive viewing	SSQ	Correlates	EEG across speed, complexity, and stereoscopy
[100]	EEG	9	Screen	Active navigation	SSQ	Classification	Rule-based EEG severity classification
[101]	GVS+EEG	19	HMD	Active navigation	SSQ	Mitigation	Motion-synchronised GVS during flight VR
[102]	GVS+posturography	140	HMD	Passive viewing	custom	Mitigation	GVS probing of vestibular contribution
[103]	EEG+EMG	28	HMD	Passive viewing	MSSQ	Classification	BioVRSea EEG+EMG sickness prediction
[104]	EEG+EOG	20	HMD	Passive viewing	FMS	Classification	EEG preprocessing trade-offs for cybersickness
[105]	EEG+GSR	22	HMD	Passive viewing	SSQ	Classification	Explainable AI on multimodal physiology
[106]	EEG	25	HMD	Passive viewing	SSQ	Classification	PLV connectivity CNN-LSTM for VIMS
[107]	GVS+GSR	20	HMD	Passive viewing	SSQ	Mitigation	Wearable motion-synchronised GVS
[108]	EEG	22	HMD	Passive viewing	SSQ	Correlates	EEG comparison of HMD versus screen VR
[109]	EEG	31	HMD	Passive viewing	SSQ	Correlates	EEG across VR motion parameter variations
[110]	tDCS+ECG+posturography	20	HMD	Passive viewing	SSQ	Mitigation	Anodal tDCS over right TPJ
[39]	fNIRS+ECG+posturography	11	HMD	Passive viewing	behav.	Correlates	fNIRS and posturography for 3D sickness
[111]	EEG+ECG+EGG	28	HMD	Mixed	SSQ	Indiv. diff.	EEG aperiodic activity for susceptibility profiling
[112]	EEG	53	HMD	Passive viewing	SSQ	Mitigation	Real-time EEG-driven factor-specific mitigation
[113]	EEG	6	HMD	Active navigation	SSQ	Mitigation	Gaze-contingent peripheral blur mitigation
[114]	Other	108	Projector	Mixed	SSQ	Mitigation	Bone-conducted vibration vestibular stimulation
[115]	GVS	40	HMD	Active navigation	SSQ	Mitigation	Noisy GVS for nauseogenic VR
[116]	EEG	6	Projector	Driving (passive)	cont.	Classification	Continuous EEG-based motion-sickness estimation
[117]	EEG	40	TV	Passive viewing	SSQ	Recovery	EEG recovery dynamics after VR exposure
[118]	EEG	17	HMD	Active navigation	SSQ	Cognition	Inhibition-related ERPs after VIMS
[119]	EEG	12	HMD	Other	SSQ	Cognition	Full-body VR exergame and EEG theta
[120]	EEG	16	HMD	Passive viewing	SSQ	Classification	Squeeze-excitation CNN for severity regression
[38]	fNIRS	1	HMD	Active navigation	SSQ	Mitigation	fNIRS-triggered FOV-narrowing mitigation
[37]	fNIRS	10	HMD	Mixed	SSQ	Mitigation	Wearable fNIRS for adaptive FOV restriction
[121]	EEG+ECG	64	HMD	Passive viewing	SSQ	Classification	Interpretable spiking-network EEG prediction
[122]	EEG	25	HMD	Mixed	SSQ	Mitigation	Multisensory mitigation with EEG correlates
[36]	fNIRS	30	HMD	Passive viewing	SSQ	Correlates	fNIRS angular-gyrus correlates of disorientation
[123]	EEG+EOG	15	HMD	Passive viewing	none	Classification	S-transform EEG features for cybersickness

A. Corpus Characteristics

The 92 included studies cluster on EEG-based investigations of young adults in head-mounted VR, but vary widely on baseline condition, montage, and preprocessing. The dispersion documented in this subsection is the precondition for the meta-analytic observations reported in subsection IV-F.

1) *Sample Sizes and Demographics*: The 92 studies span 2001–2025. Across the corpus, 2687 participants were analysed (recruited: 2822), with a mean of 29.2 ± 29.0 participants analysed per study (median: 22, range: 1–202). The 62 studies that report a parsable adult-age range yield a mean age of 23.7 ± 2.7 years, indicating a focus on young-adult populations; 14 studies do not report age or report ranges that cannot be parsed (Figure 3).

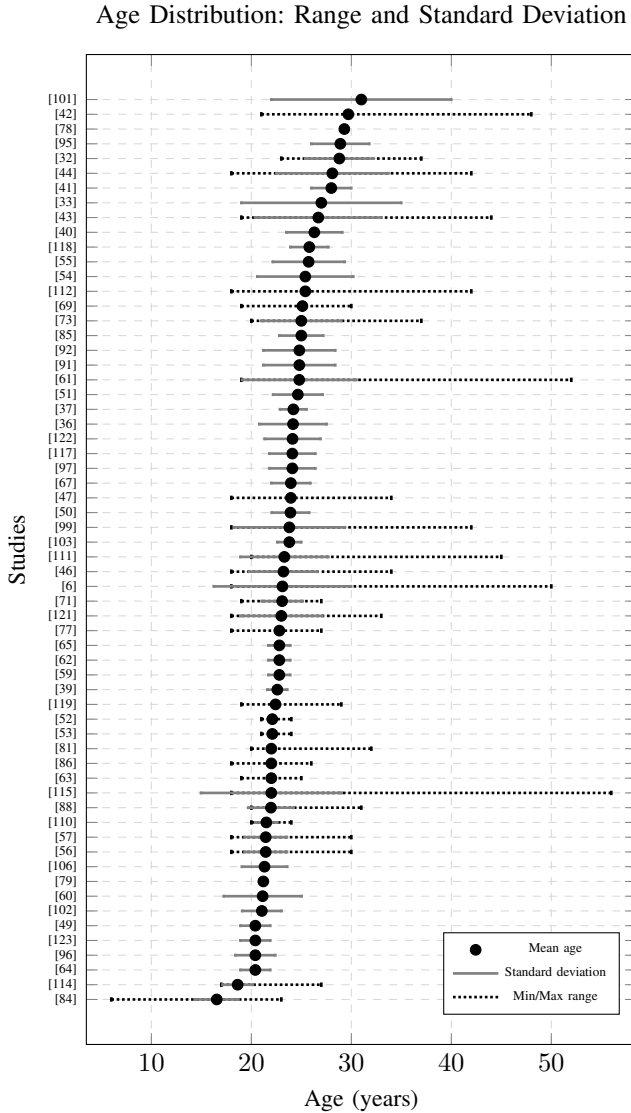


Fig. 3: Age distribution across studies showing mean age (black dots) with age ranges (dotted black error bars) and standard deviations when available (solid gray error bars). The age range shows the minimum and maximum participant ages, while the standard deviation shows the typical spread around the mean age.

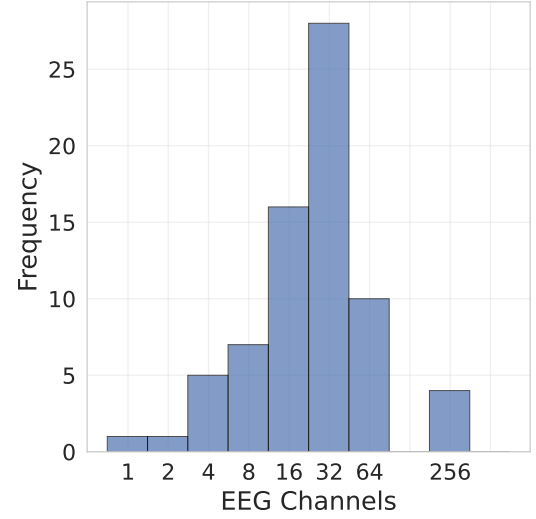


Fig. 4: Distribution of EEG channel configurations across cybersickness studies.

2) *Neuroimaging and Physiological Recording Modalities*: EEG is the predominant modality (77 of 92 studies, 83.7%). The remaining studies use fNIRS (4 of 92, 4.3%) or non-invasive neurostimulation without concurrent recording (11 of 92, 12.0%: GVS in 5, tDCS in 3, tACS in 2, and other in 1). No corpus study uses fMRI, MEG, or PET; section III states the search-query and screening implications of this absence.

EEG channel counts vary widely (median 30, range 1–256, mean 39.6 ± 55.8 , $n = 72$ studies reporting). The most frequent montages are 32-channel (18 studies), 64-channel (9), 14-channel (8), 16-channel (5), and 30- or 256-channel (4 each); a long tail of low-channel configurations (4 to 16 channels) is consistent with portable in-VR recording (Figure 4). Channel count is positively correlated with reported within-subject classification accuracy ($r = 0.403$, $p = 0.037$, $n = 27$); this observation is revisited in subsection IV-F.

Concurrent physiological recording is reported in 36 of 92 studies (39.1%), with the criterion restricted to physiological sensors (electrocardiography, galvanic skin response, electro-oculography, electromyography, electrogastrography, photoplethysmography, respiration, skin temperature, blood pressure). The most common adjuncts are electro-oculography (18 studies), electrocardiography (16), galvanic skin response (9), photoplethysmography (7), and electrogastrography and skin temperature (4 each). Other co-recorded signals (head tracking, joystick, button input) are listed separately in the all-studies table.

3) *Experimental Design and Data Collection Protocols*: Study designs predominantly employ within-subject comparisons (78 of 92, 84.8%), with 7.6% employing between-subject designs exclusively and 7.6% using both. The majority of protocols are single-session (mean 1.6 ± 2.1 sessions, median 1, $n = 91$). Session duration averages 36.3 ± 32.6 minutes (median 25.8, range 1.8–180, $n = 76$), and the cybersickness-inducing stimulus block within a session averages 15.2 ± 15.4 minutes (median 10.0, range 0.5–90, $n = 86$).

Virtual reality is delivered through head-mounted displays

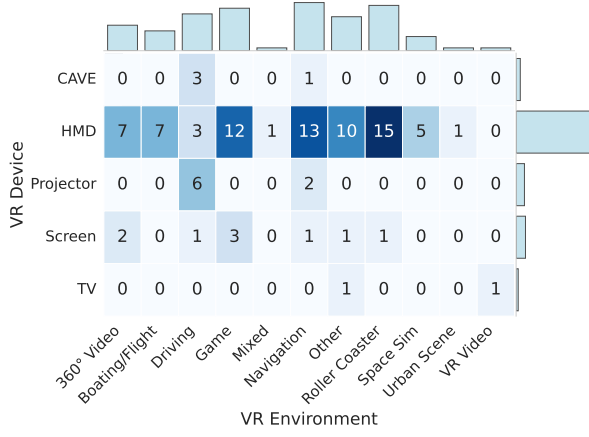


Fig. 5: Heatmap of the distribution of VR device types and virtual environments across studies. Studies employing multiple devices or environments contribute towards each.

in 73.9% of studies (68 of 92), with screens (8.7%) and projection systems (8.7%) as the next most common configurations, followed by CAVE (4.3%), television (2.2%), and other or mixed-modality setups (2.2% combined). Virtual environments cluster on navigation (17 studies, any-mention 18.5%), roller-coaster experiences (16, 17.4%), games (15, 16.3%), driving (13, 14.1%), passive video (10, 10.9%), boating or flight (7, 7.6%), and space simulators (5, 5.4%). The joint distribution of device type and environment is shown in Figure 5; the hardware mix bears on the meta-analytic findings in subsection IV-F.

4) *Cybersickness Assessment Methods*: Cybersickness is measured through questionnaire instruments in 70 of 92 studies (76.1%), continuous ratings during exposure in 20 (21.7%), single-item Likert scales in 15 (16.3%), and behavioural indices in 8 (8.7%); these categories sum above 100% because 19 studies use more than one method on the same exposure. Among studies using questionnaires, the Simulator Sickness Questionnaire is the most common instrument (59 of 92), followed by the Fast Motion Sickness Scale (6), the Motion Sickness Susceptibility Questionnaire (3), the Virtual Reality Sickness Questionnaire (2), and other instruments (6); 16 studies do not specify the questionnaire used. The use of multiple measurement methods on the same exposure indicates that the operational definition of cybersickness varies across the corpus, which constrains cross-study comparability.

5) *Baseline Conditions and Preprocessing Heterogeneity*: Baseline conditions vary across the corpus: 39 of 92 studies use a task baseline, 16 use an eyes-open resting baseline, 8 use a pre-stimulus segment of the same recording, 5 use an eyes-closed resting baseline, and the remainder either do not specify a baseline (3) or do not report one (20 missing or not applicable, including the neurostimulation-only studies). The five eyes-closed cases are a minority of the corpus, but alpha-band reactivity to eye closure overlaps with the alpha modulations attributed to cybersickness; section V develops this critique and discusses recommended alternatives.

EEG signal preprocessing is also heterogeneous in filter

selection. Band-pass filtering ranges from very low frequencies (0.01 Hz) to high gamma (100 Hz), with the most frequent configurations being 1–50 Hz (9 studies), 0.5–50 Hz (7), 0.5–100 Hz (4), and 0.5–45 Hz (4); 25 studies do not specify a band-pass filter. A 50 or 60 Hz notch filter is applied in 18 studies; 72 studies do not specify a notch filter. The full distribution of band-pass configurations is shown in Figure 6.

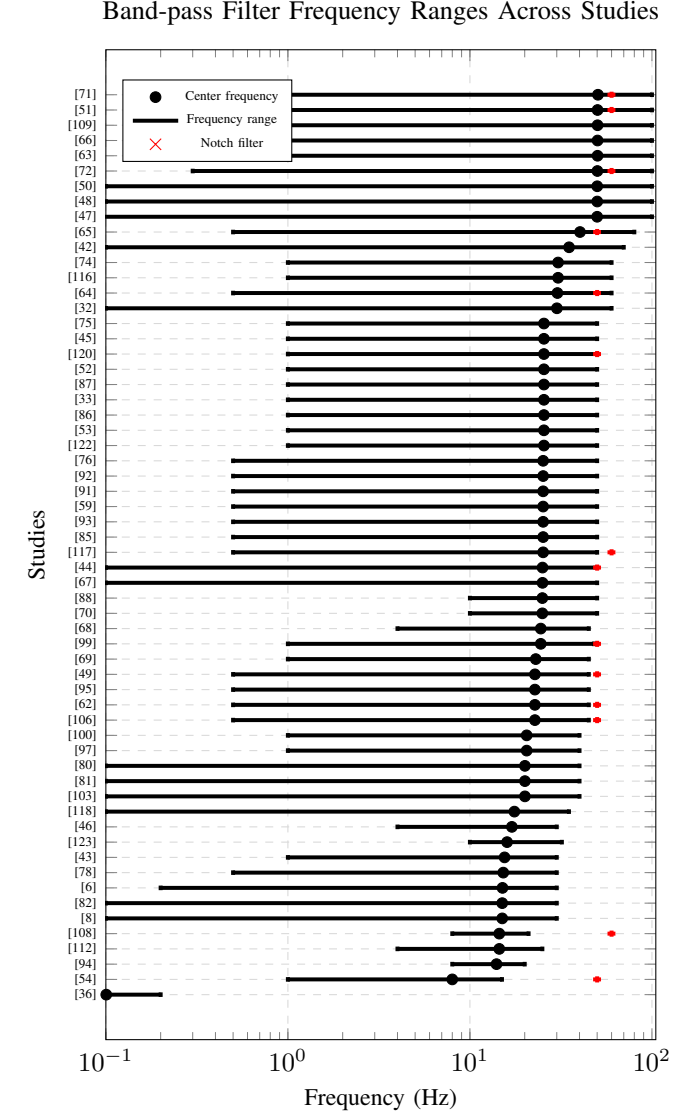


Fig. 6: Band-pass filter frequency ranges across studies showing center frequency (black dots) with frequency ranges (black error bars) and notch filter frequencies (red crosses). The frequency range shows the low and high cutoff frequencies of the band-pass filters, while red crosses indicate notch filter frequencies used to remove specific interference (e.g., 50/60 Hz power line noise).

B. Neural and Physiological Correlates of Cybersickness

Across the 77 EEG, 4 fNIRS, and a subset of corpus studies using event-related potential analyses, the corpus converges on a low-frequency power increase during cybersickness; higher-frequency and lateralised findings are less consistent at the

corpus level and, as shown in subsection IV-F, are partly attributable to differences in VR hardware platform.

1) *EEG Spectral Power*: Across the studies reporting per-band changes during cybersickness, the most consistent direction is an increase in low-frequency power. Of the 30 studies reporting a delta-band change, 16 (53.3%) report an increase, 6 a mixed pattern, 5 no change, and 3 a decrease. Of the 35 studies reporting a theta-band change, 20 (57.1%) report an increase, 6 a decrease, 5 a mixed pattern, and 4 no change. Increases in delta (1–4 Hz) and theta (4–8 Hz) power have been associated with the onset and increasing severity of cybersickness across diverse VR stimuli and hardware [32], [33], [43], [67], [71].

Alpha-band findings diverge across the corpus. Of the 39 studies reporting an alpha change, 15 (38.5%) report an increase, 11 (28.2%) report a decrease, 9 (23.1%) report a mixed pattern, and 4 (10.3%) report no change. Studies that report alpha suppression, particularly over central, parietal, and occipital cortices, interpret it as a marker of increased cognitive load under sensory conflict or as a consequence of postural-control demand [26], [53], [70]. Studies that report alpha enhancement attribute it to sensory gating or to drowsiness during prolonged exposure. subsection IV-F shows that the direction of the alpha change tracks VR display hardware, and reframes part of this corpus-level heterogeneity as hardware-attributable.

Higher-frequency bands show less consistent patterns than delta and theta. Of the 36 studies reporting a beta-band change, 11 (30.6%) report an increase, 10 (27.8%) a decrease, 10 a mixed pattern, and 5 no change. Individual reports range from beta suppression [71] to beta enhancement [109], and one study identifies beta as the most informative band for channel selection in classification [124]. Of the 21 studies reporting a gamma-band change, 7 (33.3%) report an increase, 6 no change, 5 a mixed pattern, and 3 a decrease. The full distribution of rhythm changes across anatomical regions is shown in Figure 7; as with alpha, the beta result is partly attributable to display hardware (subsection IV-F).

The test-retest reliability of these spectral markers has been assessed in one study, which reported good reliability for delta, theta, and alpha power changes in frontal and central areas across multiple sessions [85]. Delta-band normalisation has also been used as an objective marker of the post-exposure recovery time course [117]; the recovery literature is discussed further in section V.

Beyond band-power, a subset of studies has examined complex spectral and information-theoretic features of the EEG signal, including spectral entropy and skew [59], Gravity Frequency and Kolmogorov Complexity [89], [90], amplitude-modulation features [104], and the mean frequency of the theta band as a direct sickness index [125]. Functional connectivity has also been investigated, although reporting is sparse: of the 16 studies that report a connectivity change, 7 report a mixed pattern, 4 a decrease, 3 an increase, and 2 no change. Specific findings include a decrease in information transfer measured via transfer entropy with the onset of VIMS [32], microstate-level alterations of canonical brain networks and decreased connectivity between visual cortex (V5) and

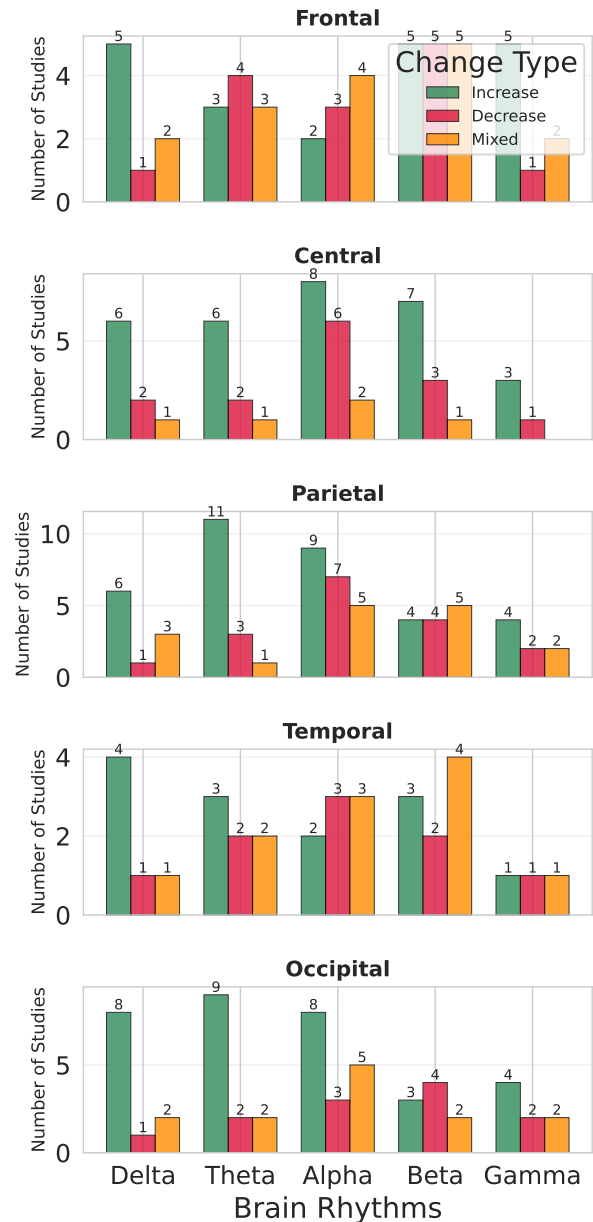


Fig. 7: Brain rhythm changes across anatomical regions during cybersickness.

vestibular processing regions [97], and phase-locking-value features that serve as inputs to deep classification models [66], [106]. The causal relevance of these spectral and connectivity patterns is suggested by intervention studies in which symptom mitigation correlates with a normalisation of the same EEG bands [122], [126].

2) *Event-Related Potentials and Evoked Responses*: Beyond ongoing oscillations, a subset of studies has used event-related potentials (ERPs) to characterise how cybersickness affects discrete cognitive processes. These studies indicate that cybersickness is accompanied by measurable changes in attentional allocation and response inhibition. Cybersickness severity is associated with a reduction in the amplitude of the P3b component in a dual-task paradigm, a classic marker

of diminished attentional resources [6]. VIMS has also been shown to impair response inhibition, as reflected by a larger N2 amplitude and a smaller, delayed P3 component during an oddball task [118]. Read alongside the sensory-conflict and cognitive-load accounts introduced in section II, these findings are consistent with cybersickness consuming processing resources that would otherwise support task performance.

Susceptibility-related ERP patterns recur in studies that compare resistant and susceptible participants. VIMS-susceptible participants exhibit excessive parietal N2 amplitudes and impaired theta-band phase synchronisation compared to resistant counterparts [127], and individuals with high sickness levels show higher P3 amplitudes during sensory-mismatch conditions [42]. A separate ERP approach has linked the subjective experience of disorientation to a suppression of fronto-central heartbeat-evoked potentials, providing a link between interoceptive processing and a key symptom of cybersickness [51]. These susceptibility markers are revisited in subsection IV-D alongside the resting-state-prediction literature.

3) *Hemodynamic Responses (fNIRS)*: Four fNIRS studies in the corpus report convergent regional hemodynamics during cybersickness, providing a small but coherent body of evidence on the cortical regions involved. Bilateral angular-gyrus deactivation has been reported in association with subjective sickness scores [36], and increases in total hemoglobin concentration have been linked directly to cybersickness onset [37], [38]. Reduced cortical activation during low-resolution stereoscopic exposure has also been documented [39].

The direction of the hemodynamic effect is not uniform across the small fNIRS literature: regional perfusion increases in some cortical areas coexist with decreases in others, and fNIRS-derived hemoglobin signals have been used as input features for regression models predicting cybersickness severity [128]. Yamamura et al. also demonstrate the only fNIRS-based closed-loop mitigation system in the corpus, which is taken up in subsection IV-E.

C. Classification and Prediction of Cybersickness

Of the 92 corpus studies, 38 have classification or prediction of cybersickness as their primary objective; a broader 41 use classification or regression methods as part of mixed-objective work. Corpus-level claims in this subsection are restricted to the 38-study primary-objective subset. Across these 38 studies, accuracy is the dominant reported performance metric, but sensitivity/specificity, AUC/F1, and the operational definition of the sick/non-sick label are reported in fewer than half of the subset.

1) *Methodological Approaches and Feature Engineering*: Computational approaches across the 38-study subset divide between traditional machine learning (15 of 38, 39.5%), deep learning (14, 36.8%), and regression (6, 15.8%); thresholding (1) and other approaches (2) account for the remainder. A subset of these studies uses multimodal pipelines that combine EEG with physiological signals or with VR content features.

Traditional machine-learning pipelines rely on hand-crafted features, with Support Vector Machines, k-Nearest Neighbours, and Linear Discriminant Analysis as the most common

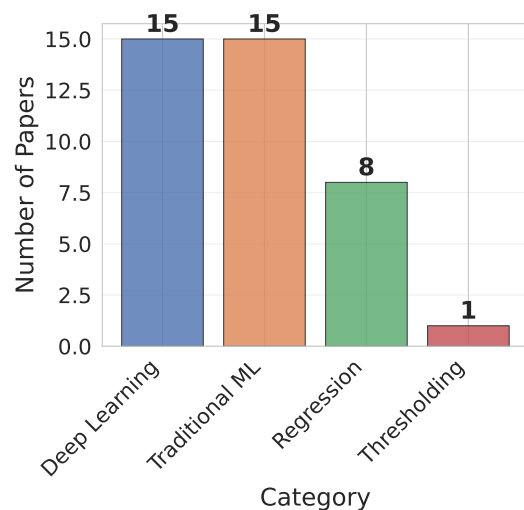


Fig. 8: Distribution of classification method categories across cybersickness studies.

classifiers [49], [129]. Features derived from EEG include Power Spectral Density, wavelet-derived energy ratios [78], spectral entropy and skew [59], and time-domain statistics such as signal variance [94]. Dimensionality reduction is addressed through feature selection algorithms such as Correlation Feature Selection [124] and genetic algorithms [130].

Deep-learning pipelines learn task-relevant features directly from the data. Convolutional Neural Networks (CNNs) are widely used to extract spatial and spectral features from EEG time series, spectrograms, or topographic scalp maps [68]. Temporal dynamics are captured through recurrent layers, including Long Short-Term Memory [84] and Gated Recurrent Units [93]. Other architectures include spiking neural networks [121] and shallow CNNs designed for real-time detection in mitigation systems [112]. Recent models employ hybrid architectures combining CNNs with LSTMs [106], dual-pathway designs that process raw EEG and functional-connectivity inputs in parallel [66], and attention mechanisms that weight the contribution of different channels or time points [63]. Inputs to these models have diversified to include functional-connectivity matrices [106] and cross-frequency coupling asymmetries [64].

Multimodal fusion combines EEG with peripheral physiological signals such as galvanic skin response and blood volume pulse [105], or with non-neural signals such as electromyography from postural muscles, which has been reported as more informative than concurrently recorded EEG in one study [103]. Other approaches use fNIRS as a regression input [128] or transfer knowledge from EEG-trained models to enable prediction from VR video content alone [72]. Explainable-AI techniques have been incorporated in a subset of studies to interpret model decisions [105].

2) *Reported Performance and Discrimination Metrics*: Model validation predominantly employs k-fold cross-validation (21 of 38, 55.3%), with train-test splits (6), leave-one-subject-out cross-validation (1), and other or unspecified schemes (10) making up the remainder. The distribution of

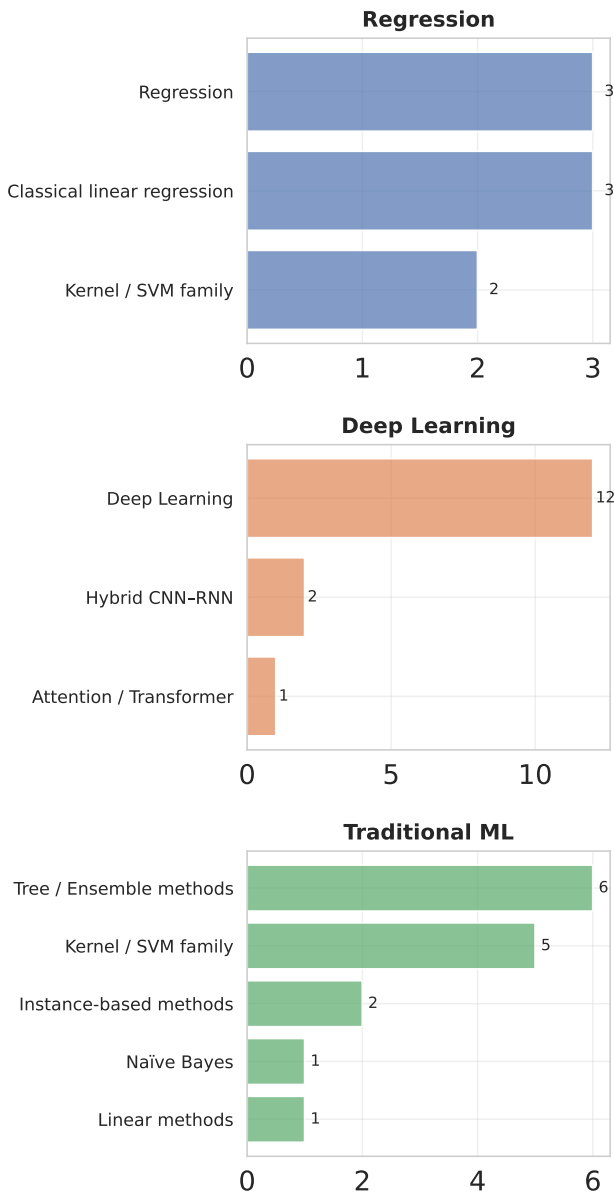


Fig. 9: Detailed breakdown of specific methods within each classification category.

methods and validation schemes is summarised in Figure 8 and Figure 9.

Reported within-subject classification accuracies are high but rest on a small denominator: across the 8 of 38 studies that report a within-subject accuracy, the mean is $90.7\% \pm 8.4\%$ (median 91.5%, range 73.3–100.0%). Cross-subject accuracies are reported by 22 of 38 studies, with a mean of $88.1\% \pm 11.6\%$ (median 88.9%, range 55.0–99.7%). The two distributions are compared in Figure 10, and the temporal trend of reported accuracies is shown in Figure 11.

These headline accuracies are reported on heterogeneous discrimination metrics. Of the 38 classification studies, 13 (34.2%) report accuracy as the sole discrimination metric; 7 (18.4%) report sensitivity and specificity; 10 (26.3%) report AUC or F1; the remaining 8 report only regression metrics

(R^2 , RMSE) or no quantitative performance. This is an observational corpus-level fact; the interpretive critique — that accuracy is uninformative under class imbalance and arbitrary thresholds — is taken up in section V.

3) *Classification Tasks*: Across the 38-study subset, binary classification of a “sick” versus “not sick” state is the most common task. Several studies report within-subject accuracies above 95% on this binary task [57], [63], [64], [68], [106], [129]. The sick/non-sick label that grounds these accuracies varies in how it is operationalised: 19 of 38 classification studies expose the threshold or its basis, of which 8 use a numeric cutoff and the remainder use self-reported binary labels, pre/post differences, total-questionnaire-score cutoffs, or do not specify; the other 19 studies do not document the threshold at all.

A second task is multi-level classification, which categorises cybersickness into several discrete levels of severity. Three-level classification has been demonstrated with multimodal fusion [8], [77] and with EEG alone using rule-based induction [100] or a genetic algorithm combined with a Support Vector Machine [87].

A third task is continuous prediction or regression of a cybersickness severity score. Performance is measured by correlation coefficients and error metrics. Deep-learning models have demonstrated high correlations between predicted and actual SSQ scores [62], [91], [92], [120]; earlier work used Support Vector Regression [74], [116] and linear regression [131]. Current architectures include Siamese LSTMs [65] and frameworks that fuse physiological signals with VR video features [76]. Addressing the threshold-heterogeneity problem raised above, one recent study develops a method for continuous, calibration-free detection directly from physiological data with dynamic, real-time ground-truth labels [132].

D. Individual Differences and Susceptibility

Resting-state neural features have been used to predict cybersickness susceptibility before exposure, and during exposure susceptible and resistant individuals show distinct neural signatures. Demographic moderators are reported in a small subset of the corpus and warrant cautious interpretation given the small effect sizes documented in the broader literature.

Pre-exposure prediction of cybersickness has been demonstrated in three corpus studies. A spiking neural network classified subsequent sickness with 85.9% accuracy from pre-exposure resting-state EEG alone [121]. Resting-state EEG beta power in the left parietal region has been reported as a significant predictor of subsequent motion-sickness ratings in an in-car VR platform [82], and aperiodic components of the EEG signal have been reported as predictive of susceptibility to rotational motion [111]. These findings point to pre-existing neural traits that influence susceptibility, although the evidence base remains small.

During cybersickness exposure, resistant and susceptible individuals show distinct neural signatures. Resistant individuals exhibit enhanced theta activity in the left parietal cortex, a candidate biomarker of resilience [81]. Susceptible individuals show excessive parietal N2 ERP amplitudes and

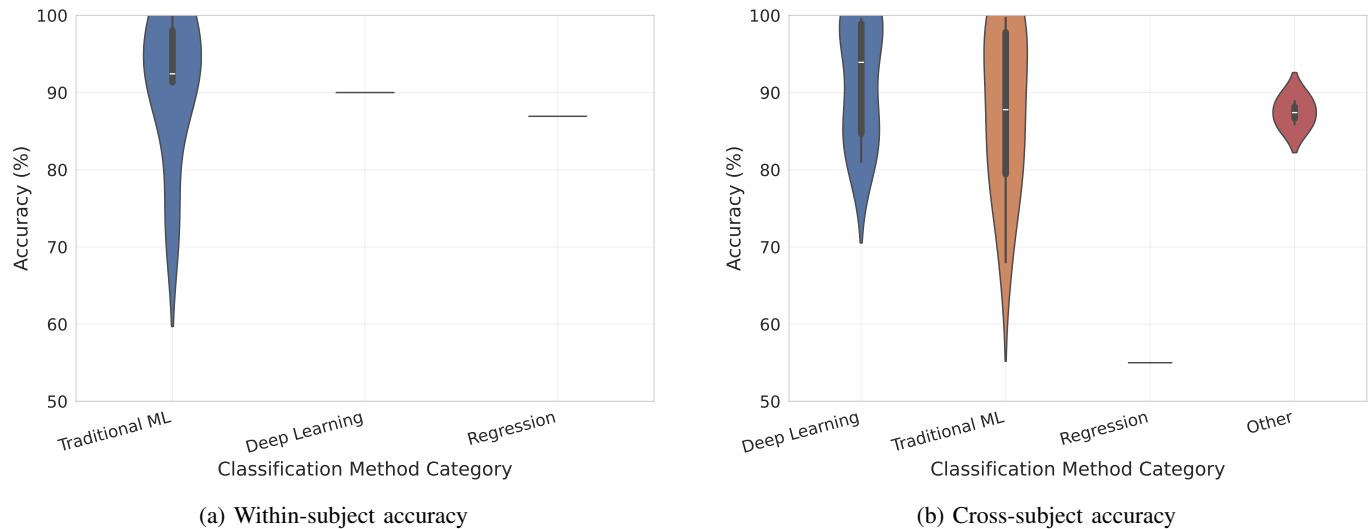


Fig. 10: Classification accuracy distributions by method category for within-subject and cross-subject validation approaches.

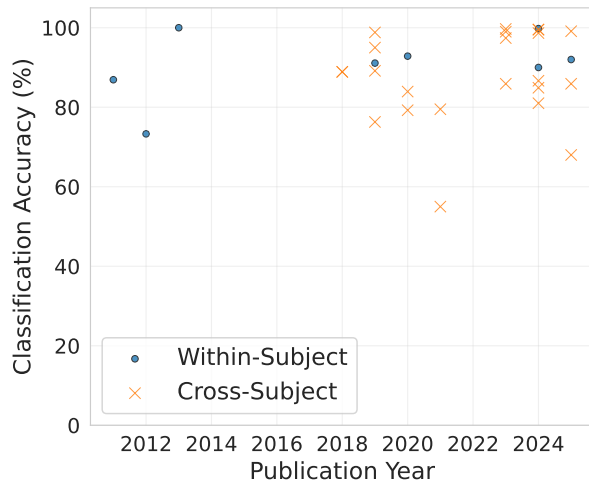


Fig. 11: Classification accuracy over time, comparing within-subject and cross-subject validation approaches.

impaired theta-band phase synchronisation [127], higher P3 amplitudes during sensory-mismatch conditions [42], and differential spectral and source-space activation patterns duringvection [133]. One study reports that the cybersickness condition produces increased delta and decreased alpha power in both susceptible and non-susceptible groups, but susceptible participants additionally show lower absolute power in the beta and gamma bands at baseline [67]. An earlier study using a long-duration driving task found that the ratio of theta to total EEG power in frontal and parietal lobes differed between sick and non-sick groups, proposing this ratio as a physiological index of sensitivity [134].

Sex and gender differences are reported in 3 of 92 corpus studies, all from a single laboratory using a VR-neurofeedback paradigm [46]–[48]; an additional 11 corpus studies use male-only samples, and the remaining 78 do not analyse sex effects. The broader cybersickness literature outside the corpus calibrates these findings. A recent meta-analytic synthesis con-

cludes that women experience greater cybersickness than men, but that the effect size is modest (meta-analytic $r \approx 0.21$) and that the field is widely underpowered, requiring approximately 176 participants per arm for 80% statistical power to reliably detect the effect [135]. A separate review of 76 studies finds that only 37 of those studies (48.7%) are consistent with the assertion that women are more susceptible than men [136]. In a single experimental study, the difference between male and female participants did not reach significance when gaming experience was included as a covariate [137]. Read alongside this broader literature, the 3 of 92 corpus reports are consistent with an underpowered signal of a small effect that gaming experience and other moderators may partially explain, rather than with a corpus-wide sex difference.

Individual variability in cybersickness susceptibility has measurable neural correlates and small demographic moderators. At present, the corpus is underpowered for either to be operationalised as a predictive biomarker in a deployment setting. The mitigation literature reviewed next acts on the neural signals identified here.

E. Mitigation and Intervention Strategies

Mitigation research in the corpus splits into two threads: closed-loop adaptive VR that modifies the virtual environment in response to detected sickness (2 corpus studies), and interventions that target the user's brain state directly (5 corpus studies including transcutaneous auricular vagus nerve stimulation, vibro-motor reprocessing therapy, multisensory cueing, and neurofeedback). Both threads are early-stage, and the small literature already documents one methodological obstacle specific to the closed-loop setting.

1) *Closed-Loop Adaptive VR*: Only two corpus studies implement closed-loop adaptation: the literature is in early-stage development. One system uses a two-stage shallow convolutional network to detect cybersickness from EEG and infer a likely causal factor, then selectively modifies the implicated VR parameter (navigation speed or scene complexity),

producing a significant reduction in user-reported sickness compared to a control condition [112]. A second system uses fNIRS to monitor total hemoglobin concentration; on detecting increases correlated with sickness, the system dynamically reduces the user's field of view, demonstrating the feasibility of an fNIRS-based closed-loop pipeline [37].

2) *Interventions Modulating Brain State*: Five corpus studies investigate interventions intended to act on the underlying physiological state of the user, with concurrent EEG used to verify neural engagement. Transcutaneous auricular vagus nerve stimulation significantly reduced motion-sickness symptoms in one study, with the reduction correlated with stimulation-induced neural activation in specific brain regions measured by EEG source localisation [138]. Vibro-motor reprocessing therapy was associated with significant reductions in alpha, delta, and theta spectral power [126]. Providing synchronised auditory and physical motion cues during a simulated bicycle ride reduced VIMS scores relative to conditions with mismatched or absent cues, with the effect reflected in concurrent EEG recordings [122].

Two further studies investigated neurofeedback protocols in which users were trained to volitionally modulate their own brain activity. The presence of cybersickness symptoms was found to directly impair the user's ability to successfully perform the neurofeedback training task [48], and a placebo transcranial direct current stimulation procedure — while having no significant effect on reported sickness symptoms — paradoxically degraded performance on a concurrent neurofeedback task [47]. These findings illustrate a methodological challenge for any neuromarker-driven closed-loop intervention: the physiological state being targeted can interfere with the user's ability to engage with the intervention itself.

F. Corpus-Level Methodological Observations

Meta-analytic comparison across the 92 corpus studies reveals that VR display hardware substantially modulates EEG band-power responses, that stimulus-exposure duration is confounded with hardware, and that the corpus shows signs of funnel-plot asymmetry consistent with publication bias. The findings in this subsection are observational; the interpretive critique and reporting recommendations they motivate are developed in section V.

Alpha-band cybersickness responses differ between head-mounted-display and screen-based studies. A Mann-Whitney U test on the direction of alpha change between HMD studies ($n = 24$ reporting) and screen-based studies ($n = 4$) yields Cliff's $\delta = 0.875$ ($p = 0.004$), a large effect. The underlying distribution is asymmetric: HMD studies report alpha increases four times as often as decreases (12 of 24 versus 3 of 24), whereas screen-based and other non-HMD studies report alpha decreases nearly three times as often as increases (8 of 15 versus 3 of 15). This pattern reframes the alpha-band heterogeneity reported in subsection IV-B as partly hardware-attributable, rather than as a pure intersubject or task effect.

The same comparison for the beta band yields Cliff's $\delta = 0.664$ ($p = 0.015$, large effect; HMD $n = 25$, screen $n = 5$), indicating that the hardware effect extends beyond alpha. A

Kruskal-Wallis test across the full VR-device categorisation yields $\eta^2 = 0.251$ ($p = 0.009$) for alpha and $\eta^2 = 0.160$ ($p = 0.033$) for beta, corroborating the binary hardware comparison. The hardware effect does not extend to delta, theta, or gamma, which show no significant between-device differences; the claim is therefore bounded to higher-frequency bands.

Stimulus-exposure duration is itself confounded with hardware. HMD studies use shorter exposure periods (mean 12.6 ± 14.5 minutes, $n = 65$) than non-HMD studies (mean 23.0 ± 15.7 minutes, $n = 21$); Mann-Whitney $U = 364.0$, $p = 0.001$, Cliff's $\delta = -0.467$ (medium effect). The hardware confound therefore covaries with how long the user is exposed, not only with the display optics; comparisons of band-power changes between HMD and screen studies necessarily mix the two effects.

Across all reported classification accuracies in the corpus, Egger's regression test of funnel-plot asymmetry yields $p = 0.011$, consistent with possible publication bias toward high-accuracy results. Egger's test is sensitive to between-study heterogeneity and small-study effects, and the present result should not be read as a definitive demonstration of bias; the interpretive caveats and the methodological recommendations that follow from it are developed in section V.

These corpus-level observations, together with the inline reporting-gap counts in subsection IV-C and the baseline-heterogeneity findings in subsection IV-A, seed the design and reporting recommendations developed in section V.

V. DISCUSSION

This work follows a recent scoping review by [1], which provides an overview of the use of EEG in cybersickness research. Their work, based on a literature search conducted in September 2021, systematically analyzed 33 EEG-based studies. The review reported that frequency or time-frequency analysis was the predominant method for feature extraction. Key neurophysiological findings identified in their analysis included an increase in delta power and a decrease in beta power associated with cybersickness, while results for the alpha band were noted as being inconsistent across studies. The authors also reported that classification models achieved accuracies between 79% and 100%, and that studies frequently used head-mounted displays with portable EEG systems and recruited participants primarily in their 20s. The volume of research in this domain has increased substantially since the conclusion of their search. Our analysis includes 49 additional studies published after their search date.

A. Methodological Challenges and Critical Perspective

While the reviewed literature demonstrates progress in applying neuroimaging and neurostimulation methods to cybersickness research, a critical appraisal reveals substantial methodological challenges. These issues temper the interpretation of many findings and present considerable barriers to the generalization, replication, and practical application of the reported results. This section will highlight four main challenges identified in the literature (1) optimistic accuracy estimates, (2)

the lack of standardized cybersickness thresholds, (3) protocol disparities and (4) conceptual variability.

A prominent trend in the classification literature is an “accuracy arms race,” with a proliferation of studies reporting exceptionally high performance. Numerous papers claim binary classification accuracies exceeding 95%, and in some cases, approaching 99% [57], [63], [64], [66], [106]. However, these impressive figures are often a product of experimental designs that limit their generalizability. Many are achieved using within-subject validation on small, homogeneous participant pools, which may not translate to new, unseen users. Furthermore, the choice of data used for classification also plays a role. For example, some high-performing models compare resting-state EEG data collected before and after VR exposure [49], [63], a paradigm that, while useful for identifying state changes, is ill-suited for the real-time detection required by adaptive systems.

The foundation of any supervised learning model is the quality of its “ground truth” labels, and in cybersickness research, this foundation is ill-defined. The vast majority of predictive models are trained on subjective data, most commonly post-exposure SSQ scores or in-task ratings from scales like the FMS [104]. The SSQ is retrospective and coarse, while continuous ratings can lead the model to classify signals related to the motor action of reporting. A more fundamental issue is the lack of standardized thresholds for defining sickness states. Individual studies frequently devise their own arbitrary cut-offs to convert continuous SSQ scores into binary (“sick” vs. “not sick”) or multi-level (“mild,” “moderate,” “severe”) labels. This means that the very definition of the target variable is inconsistent across the literature, severely undermining the comparability of different models and their reported accuracies.

This lack of standardization extends to many aspects of experimental design, creating a profound heterogeneity that hinders scientific synthesis. First, there is significant variability in display technology and the nature of the stimulus. Experiences in fully immersive HMDs are not equivalent to those on large-screen monitors or in CAVE-like projection systems [85], [139], or smartphone based VR systems [126], yet findings are often discussed interchangeably. Similarly, passive viewing of a 360-degree video [109] is fundamentally different from active navigation in an interactive game [26]. Our meta-analysis supports these concerns, revealing statistically significant differences in neural response patterns between different hardware platforms and virtual environment types (see subsection IV-F). Second, experimental protocols are inconsistent, with wide variance in exposure duration, the inclusion and length of rest periods, and the definition of a valid baseline. A particularly problematic practice is the use of an “eyes-closed” baseline, which artificially inflates alpha power and directly confounds the interpretation of alpha suppression, one of the most commonly reported neural markers of cybersickness. Third, the recording itself varies, ranging from research-grade, high-density EEG systems to low-cost, consumer-grade devices with fewer channels and potentially lower signal quality, which may impact the reliability of the derived results [90]. Additionally, reporting quality

across studies demonstrates substantial deficiencies in critical methodological parameters. For example, EEG reference electrode configuration is reported in only 49.4% of studies, and notch filter frequency in 22.9%, hampering reproducibility and meta-analytic synthesis. A quantitative analysis of the included literature reveals a reliance on low-density EEG systems, with 82.9% of studies employing 32 or fewer electrodes, and 21.4% employing 10 or fewer electrodes. While such systems are appropriate for applied BCI paradigms that target well-characterized and localized neural signals, their use in the context of cybersickness research presents a significant methodological limitation. The neural underpinnings of cybersickness are not yet fully elucidated, and the research objective for many studies remains fundamentally exploratory. Low-density electrode montages have insufficient spatial resolution for reliable source localization and are blind to activity originating from large, un-sampled cortical areas. This practice, therefore, risks generating an incomplete understanding of cybersickness neurophysiology, potentially overlooking critical contributions from distributed neural networks and biasing the literature towards the few cortical sites that are consistently measured. Compounding these protocol issues, the literature is dominated by small and demographically narrow samples (typically young, predominantly male university students), which further limits the statistical power and external validity of the reported findings [67], [126].

Finally, the field exhibits conceptual variability in both the use of theory and the use of terminology. Many studies cite an etiological theory of cybersickness in their introduction or methods, but comparatively few return to that framework when interpreting their neurophysiological findings. The consequence is interpretive ambiguity: a neural feature observed in a given paradigm could reflect sensory conflict, postural control, expectation-related vection, display optics, or some combination thereof, depending on the exposure context, and without an explicit theoretical commitment the literature cannot adjudicate between these possibilities. Terminological practice compounds the problem. Motion sickness, VIMS, simulator sickness, and cybersickness are related but non-identical constructs, each tied to a different exposure context (see section II), yet they are frequently used interchangeably across studies [11], [12], [14], [15]. A similar conflation arises between sickness symptoms and the perceptual phenomenon of vection, despite the latter being absent from the SSQ and from the Bárány diagnostic criteria [7], [12], [17]. A further consequence of this conceptual looseness is the tendency to label any statistically significant feature a “neuromarker” without establishing its specificity, sensitivity, or reliability across contexts.

Taken together, this landscape of methodological and conceptual variability poses a significant threat to replicability and impedes the cumulative progress of the field.

B. Future Directions

The successful identification of cybersickness correlates and the development of high-performing classifiers mark a significant first phase of research. However, to transition

from a collection of isolated, proof-of-concept studies to a mature scientific field capable of producing robust, real-world applications, a concerted effort toward methodological rigor and conceptual clarity is required. Based on the challenges identified in this review, we propose the following roadmap for future research.

First, the field must embrace standardization and open science. The profound heterogeneity in stimuli, hardware, and protocols is the single greatest impediment to comparing results and building cumulative knowledge. We advocate for a community-led effort to develop standardized experimental protocols and benchmark datasets. These datasets should encompass a variety of common VR experiences (e.g., passive roller coasters, active navigation) and include high-quality, multimodal data (e.g., EEG, EOG, ECG, head-motion) alongside well-defined subjective labels. The public release of such resources, as pioneered by a few research groups [56], [72], [132], is essential for validating new methods and ensuring fair, direct comparisons between different classification and prediction models. Accompanying this should be a commitment to open-sourcing analysis code and pre-trained models to enhance transparency and facilitate replication.

Second, research must move beyond correlation to causal mechanisms. While the literature has established strong correlations between specific neural patterns and cybersickness, the causal nature of these relationships remains largely unexplored. Future work should employ interventional techniques to probe these connections. For instance, non-invasive brain stimulation methods like transcranial alternating current stimulation (tACS) could be used to exogenously modulate specific neural oscillations (e.g., alpha or theta power) to determine if this directly influences sickness susceptibility or severity. Such studies, when combined with sophisticated computational neuroscience models based on frameworks like predictive coding, can help disentangle epiphenomenal correlates from the core causal drivers of the cybersickness experience.

Third, for BCI systems to be practical, they must be generalizable and validated in ecologically valid contexts. The performance of many current classifiers drops significantly in cross-subject paradigms. Future research should prioritize the development of advanced machine learning techniques, such as transfer learning and domain adaptation, to create models that are robust to inter-subject and inter-session variability without requiring burdensome user-specific calibration. Furthermore, validation must move beyond the laboratory. Studies conducted in more naturalistic, uncontrolled settings, such as the at-home gaming paradigm explored by Moineau et al. [140], are crucial for assessing the real-world viability of these systems.

Fourth, there is a critical need to innovate on the measurement of ground truth. The field's heavy reliance on subjective questionnaires is a persistent limitation. Future work should focus on developing and validating objective, implicit behavioral markers of cybersickness that can supplement or, in some cases, replace subjective reports. These could include metrics derived from user behavior within the VR environment, such as changes in task performance, head motion dynamics (e.g., freezing or reduced exploration), or eye-tracking metrics (e.g.,

blink rate, saccadic velocity). Integrating these behavioral measures with neurophysiological signals could provide a more holistic and objective picture of the user's state.

Finally, the field should adopt longitudinal study designs to investigate adaptation and recovery. The vast majority of studies are single-session experiments, leaving the neurodynamics of how users adapt to VR over time, or "get their VR legs", largely unknown. Multi-session, longitudinal studies are needed to track the neural changes that accompany this adaptation process. Similarly, while a handful of studies have examined the recovery period immediately following exposure [117], [141], a deeper understanding of the neural mechanisms governing the dissipation of symptoms is required to develop effective post-exposure recovery strategies.

VI. CONCLUSION

The adverse effects of cybersickness remain a significant obstacle to the widespread adoption and prolonged use of Virtual Reality. Brain-Computer Interfaces have emerged as a promising approach for moving beyond subjective reports toward more objective measurement, prediction, and potential mitigation of this debilitating condition. This paper has provided a systematic and critical review of the literature at the intersection of BCI and cybersickness, synthesizing the current state of knowledge and identifying key challenges and opportunities.

Our synthesis suggests an emerging neurophysiological signature of cybersickness, primarily characterized by increased low-frequency (delta, theta) power and a suppression of alpha-band oscillations in the EEG. Spatially precise modalities like fMRI and fNIRS have localized these dynamics to a network of brain regions including the insula, cingulate cortex, and areas involved in visual-vestibular processing. This foundational knowledge has facilitated the development of computational models, many based on deep learning, that show promise in classifying sickness states, though accuracy claims require careful interpretation given methodological limitations. This research has led to the development of early-stage, real-time, closed-loop adaptive systems that attempt to use neural feedback to modify the VR environment and potentially reduce user discomfort.

However, this review also highlights that the field is hampered by significant methodological heterogeneity, a lack of standardized protocols and ground-truth labels, and a frequent reliance on small, demographically-limited samples. These challenges severely limit the generalizability and replicability of many findings and impede the direct comparison of results across studies. The path toward robust, generalizable BCI applications for cybersickness management requires a concerted shift toward open science practices, including the development of benchmark datasets and standardized evaluation metrics. It also necessitates a move from purely correlational findings to causal investigations and the validation of systems in ecologically relevant, real-world contexts.

In conclusion, the integration of BCI with VR represents more than a novel tool for studying cybersickness; it is a critical enabling technology for the future of immersive

computing. The development of truly user-aware systems that can implicitly monitor and dynamically adapt to a user's neurophysiological state in real-time may help address one of the most persistent barriers to the widespread adoption of VR. This line of research is essential for paving the way for experiences that are not only more immersive and effective, but also universally comfortable and accessible.

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